LE NGUYEN GIA HUNG

AI Research Engineer · Time-Series & Multimodal XAI

heiontheway@gmail.com | +84 394 262 651 | Ho Chi Minh City, Vietnam

linkedin.com/in/le-nguyen-gia-hung | github.com/hei1sme | hei-portfolio.vercel.app

ORCID: 0009-0003-7120-8167 | Scholar: iGODAQYAAAAJ

EDUCATION

FPT University Ho Chi Minh City, Vietnam

Bachelor of Engineering in Artificial Intelligence *Expected Dec 2027 GPA: 8.62/10.0*

- **Honors:** Certificate of Merit: Honorable Student of Trimester (Semesters 3, 4, 5, 6).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence, Deep Learning, Statistical Analysis.

PUBLICATIONS & MANUSCRIPTS

Gia-Hung Nguyen Le (First & Corresponding Author) *et al.*, "Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring."
Under Review at the Australasian Joint Conference on Artificial Intelligence (AJCAI 2026).

Co-Authors, **Gia-Hung Nguyen Le** (Second Author) *et al.*, "Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution."
Under Review at the 33rd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2027).

Gia-Hung Nguyen Le (First & Corresponding Author) *et al.*, "Proactive Air Quality Forecasting and Health Alert System for Melbourne."
Australasian Joint Conference on Artificial Intelligence (AJCAI 2025). Springer, Lecture Notes in Artificial Intelligence (LNAI). *Q2 (SJR 2024)*.
DOI: 10.1007/978-981-95-4969-6_34

RESEARCH & TECHNICAL EXPERIENCE

Cyanobacterial Bloom Forecasting under Sparse Monitoring Academic Research

First Author & Corresponding Author · AJCAI 2026 Under Review *Jan 2026 – Present*

- Stress-tested multi-source time-series forecasting across 7 New Zealand lakes using a causal weekly benchmark and untouched 2023–2024 holdout ($n = 170$).
- Achieved a weighted **F1 of 0.791** for 7-day horizon forecasting; evaluated horizon-dependent fusion trade-offs and alert burden calibration.
- Developed an executable audit protocol coupling feature availability, naive baselines, incident eligibility, and paired inference contracts.

Trustworthy Multimodal Anomaly Detection in Time Series Academic Research

Second Author · ACM SIGKDD 2027 Under Review *Jan 2026 – Present*

- Developed the **Context-Conditioned Detect-Localize-Explain (C-DLE)** framework for multimodal time-series anomaly detection.
- Implemented **split-conformal calibration** on fused numerical and textual scores, delivering distribution-free control of false alarm rates.
- Designed a cross-modal counterfactual sufficiency rule achieving **94.3% attribution accuracy** against known ground truth.
- Audited temporal text leakage across 9 domains and optimized calibration pools into coresets at **10% size** without skill loss.

Proactive Air Quality Forecasting System (SmokeNet) Academic Research

First Author & Corresponding Author · AJCAI 2025 Published (LNAI Q2) *Jun 2025 – Nov 2025*

- Designed and implemented **SmokeNet**, a Transformer-based architecture for $PM_{2.5}$ time-series forecasting.
- Achieved **MAE 0.7470** and R^2 **0.9545** (35.9% error reduction vs. XGBoost); outperformed baselines by **57.7%** during severe smoke backtests.
- Engineered end-to-end pipeline processing 4+ years of data with GPU cross-validation and integrated Explainable AI (SHAP, Integrated Gradients).

MERR-GAT: Multimodal Emotion Recognition via Graph Attention Academic Research

Second Author & Corresponding Author · Targeting Elsevier Information Fusion (Q1) *Oct 2025 – Present*

- Developing **MERR-GAT**, a novel hybrid framework fusing **Wav2Vec 2.0** (audio) and **RoBERTa** (text) via Graph Attention Networks (GATv2).

- Designing a dynamic conversational graph where nodes represent utterances and edges capture temporal and speaker interactions.
- Implementing an intrinsic XAI mechanism where learned attention weights provide granular insights into the model's decision-making.
- Benchmarking on **IEMOCAP**, **RAVDESS**, and **MELD** with a dynamic classification head supporting 4, 5, or 6 emotion classes.

End-to-End Vietnamese Speech Recognition System — Academic Research
Co-Author · Speech Corpus Benchmark — *Sep 2025 – Present*

- Developing a parameter-efficient (5.4M) end-to-end ASR model using a **4-layer CNN frontend** and **3-layer Residual BiLSTM encoder**.
- Curated and trained on a massive **745-hour Vietnamese speech corpus**, optimizing with Connectionist Temporal Classification (CTC) loss.
- Achieved a **Word Error Rate (WER) of 33.09%** and **Character Error Rate (CER) of 15.28%** on diverse test sets; benchmarked against **PhoWhisper**.

LEADERSHIP

SpeedyLabX - Student Research Group — FPT University
Founder & Student Lead — *2025 – Present*

- Founded a student research group of 10 undergraduates; led weekly technical workshops and reading groups.
- Coordinated submission workflow (milestones, reviews, camera-ready) for the group's first AJCAI-accepted paper.

CERTIFICATIONS

- **Data Science Fundamentals with Python and SQL –** IBM
- **Statistics for Data Science with Python –** IBM
- **Introduction to Containers w/ Docker, Kubernetes & OpenShift –** IBM

SKILLS

- **Languages:** Python, SQL (PostgreSQL).
- **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers, PyTorch Geometric (PyG / GATv2), TensorFlow/Keras, Scikit-learn, OpenCV.
- **Explainable AI (XAI) & Evaluation:** SHAP, Integrated Gradients, Chronological CV, CTC Loss, Weights & Biases (W&B).
- **Data & MLOps/Tools:** Pandas, NumPy, Matplotlib, Git, Docker, Linux, Streamlit.
- **Technical writing:** Blog posts and technical documentation (portfolio: hei-portfolio.vercel.app).